

Time Series Analysis

TSA – ARIMA models

Materials are based on dr Piotr Wójcik TSA course (QF programme)

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DL model

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \varepsilon_t$$

$$y_t = 1 + x_t + \frac{1}{2}x_{t-1} + \varepsilon_t$$

- Let's assume the process is in a steady-state.

$$x_t = x_{t-1} = x_{t-2} = x_{t-3} = \dots = x^* = 2$$

$$y_t = y_{t-1} = y_{t-2} = y_{t-3} = \dots = y^*$$

$$\varepsilon_t = \varepsilon_{t-1} = \varepsilon_{t-2} = \varepsilon_{t-3} = \dots = 0$$

- What is the steady-state solution then?

DL model

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \varepsilon_t$$

$$y_t = 1 + x_t + \frac{1}{2} x_{t-1} + \varepsilon_t$$

- A shock happens in the period $t + 1$ and the shock is permanent. Thus

$$x_t = x_{t+1} = x_{t+2} = x_{t+3} = \dots = x^* = 3$$

$$\varepsilon_t = \varepsilon_{t+1} = \varepsilon_{t+2} = \varepsilon_{t+3} = \dots = 0$$

- What is the steady-state solution then?

$$y_N^* = \dots$$

- What is the long-term multiplier equal to?

$$LTM = \sum_{i=0}^{\infty} \beta_i$$

ARDL model

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \alpha_1 y_t + \varepsilon_t$$

$$y_t = 2 + \frac{1}{2}x_t + \frac{1}{4}x_{t-1} + \frac{1}{2}y_{t-1} + \varepsilon_t$$

- Assume a steady-state. Let's have

$$x_t = x_{t-1} = x_{t-2} = x_{t-3} = \dots = x^* = 4.$$

- In a steady-state we should have

$$y_T = y_{T+1} = y_{T+2} = y_{T+3} = \dots = y^*$$

$$\varepsilon_T = \varepsilon_{T+1} = \varepsilon_{T+2} = \varepsilon_{T+3} = \dots = 0$$

- What is the steady-state solution then?
- What is the long-term multiplier equal to?
- What would happen in case of a permanent shock in x^* ?

Outline

1. Autocorrelation and partial autocorrelation functions
 1. Autocovariance function
 2. Autocorrelation function
 3. Partial autocorrelation function
2. AR, MA and ARMA processes
 1. Definitions of AR, MA and ARMA processes
 2. Stationary conditions
 3. Properties of AR, MA and ARMA processes
3. Box-Jenkins procedure
 1. Identification
 2. Diagnostics
 3. Information criteria
4. Forecasting with AR, MA and ARMA models
 1. Forecasting with an AR model
 2. Forecasting with an MA model
 3. Forecasting with an ARMA model



Autocovariance & Autocorrelation function

Autocovariance function is given by:

- For $E[(y_t - E(y_t))(y_{t-k} - E(y_{t-k}))] = \gamma_k, \quad k = 0, 1, 2, \dots$
- Stationary processes with $E(y_t) = 0$ we have:
 - $E[y_t y_{t-k}] = \gamma_k, \quad k = 0, 1, 2, \dots$
- Autocovariance function does not have a simple interpretation, since its values depend on a scale of values of y_t .
- It gives us only information about the direction of a relationship between y_t and y_{t-k} .
- It is hard to say anything about the strength of this relationship.

Autocorrelation function

- Because of the disadvantages of the autocovariance function in practice one usually uses the autocorrelation function (ACF):
 - $\rho_k = \frac{\gamma_k}{\text{var}(y_t)}$
- Values of ACF are in the range $[-1, 1]$ so it is easy to determine the strength of a relationship between y_t and y_{t-k} .
- An important property of ACF is:
 - $\rho_k = \rho_{-k}$
 - Note also that $\rho_0 = 1$.

Partial autocorrelation function

- Partial autocorrelation function (PACF), denoted by ρ_{kk} , measures the correlation between an observation k periods ago and the current observation, after controlling for observations at intermediate lags (i.e. all lags $< k$)
- In other words, it measures the correlation between y_t and y_{t-k} , after removing the effects of $y_{t-k+1}, y_{t-k+2}, \dots, y_{t-k+1}$.
- We can imagine the difference between ACF and PACF as an analogy to the difference between regression coefficient of some regressor in simple and multiple regression model (after controlling for effects of other regressors).

AR process

- Autoregressive process (AR)

- In the autoregressive process a current value of a variable y_t , depends only on the previous values of that variable plus an error term.
- The autoregressive process of order 1, denoted as AR(1), can be expressed as:

$$y_t = \mu + \phi_1 y_{t-1} + \varepsilon_t$$

- where ε_t is a white noise disturbance term.

- If we take first p lags of y_t we get an AR(p) process:

$$\begin{aligned} y_t &= \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t = \mu + \sum_i^p \phi_i y_{t-i} + \varepsilon_t \\ &= \mu + \sum_i^p \phi_i L^i y_{t-i} + \varepsilon_t \end{aligned}$$

or

$$\phi(L)y_t = \mu + \varepsilon_t \text{ where } \phi(L) = (1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p)$$

- Such a notation is required to show some properties of an autoregressive process.

MA process

The simplest moving average process MA(1):

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

where ε_t is white noise.

- Generally, a MA(q) process is given by:

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q} = \mu + \sum_i^q \theta_i \varepsilon_{t-i} + \varepsilon_t = \mu + \sum_i^q \theta_i L^i \varepsilon_{t-i} + \varepsilon_t$$

or

$$y_t = \mu + \theta(L) \varepsilon_t \text{ where } \theta(L) = (1 - \theta_1 L - \theta_2 L^2 - \dots - \theta_q L^q)$$

- A MA model is simply a linear combination of white noise processes, so that y_t depends on the current and previous values of a white noise disturbance term.

ARMA process

- By combining the AR(p) and MA(q) models, an ARMA(p,q) process is obtained.
- In such a process the current value of some series y_t depends linearly on its own previous values plus a combination of current and previous values of a white noise error term.
- The ARMA(p,q) process could be expressed as:

$$y_t = \mu + \sum_i^p \phi_i y_{t-i} + \varepsilon_t + \sum_i^q \theta_i \varepsilon_{t-i}$$

or

$$\phi(L)y_t = \mu + \theta(L)\varepsilon_t$$

Stationarity conditions

- Stationarity conditions for an AR(p) process
 - Stationarity is a desirable property of an estimated AR model, for several reasons.
 - One important reason is that a model whose **coefficients lead to non-stationarity** will exhibit the property that **previous values of the error term** will have a **non-declining effect on the current value of y_t** as time progresses.
- The condition for testing for the stationarity of a general AR(p) process is that ALL roots of the 'characteristic equation',

$$1 - \phi_1 z - \phi_2 z^2 - \dots - \phi_p z^p = 0$$

lie outside the unit circle.

- The name "characteristic equation" comes from the fact that its roots determine characteristics of the process y_t - for example, the ACF for an AR process will depend on the roots of this characteristic equation, which is a polynomial in z .

Examples for AR process

Example 1 -- Is the following process stationary?

$$y_t = \phi_1 y_{t-1} + \varepsilon_t$$

- In order to test this, first we write y_{t-1} in lag operator notation and then take this term over to the LHS and factorise:

$$y_t = L y_t + \varepsilon_t$$

$$y_t - L y_t = +\varepsilon_t$$

$$(1 - L)y_t = +\varepsilon_t$$

- Then the characteristic equation is

$$1 - z = 0$$

having the root $z=1$, which **lies on, not outside, the unit circle**. This is why the process is called a unit root process.

Example 2 -- Is the following process stationary?

$$y_t = 3y_{t-1} + 0.25y_{t-2} - 0.75y_{t-3} + \varepsilon_t$$

- We can write the above equation as:

$$y_t = 3L y_t + 0.25L^2 y_t - 0.75L^3 y_t + \varepsilon_t$$

$$1 - 3z - 0.25z^2 + 0.75z^3 = 0$$

$$0.75(z-2)(z-\frac{1}{3})(z+2) = 0$$

Thus, the roots are: $z_1=2$, $z_2=1/3$ and $z_3=-2$. We see that only two roots lie outside the unit circle so the process is **non-stationary**.



Wold's decomposition theorem

- Wold's decomposition theorem states that **any stationary series** can be **decomposed** into the sum of two unrelated processes:
 - a purely **deterministic** part,
 - **and** a purely stochastic part, which will be an **MA(∞)**.
- A simpler way of stating this in the context of AR modelling is that any **stationary autoregressive process** of order p with **no constant** and no other terms can be expressed as an **infinite order moving average model**.
- For the AR(p) model with $\mu = 0$ and expressed as $\phi(L)y_t = \varepsilon_t$ the Wold decomposition is:

$$y_t = \psi(L)\varepsilon_t = \phi(L)^{-1}\varepsilon_t$$

Properties of an AR(p) process

- The (unconditional) mean:

$$E(y_t) = \frac{\mu}{(1 - \phi_1 - \phi_2 - \dots - \phi_p)}$$

- The autocovariances and autocorrelation functions can be obtained by solving a set of simultaneous equations known as the Yule-Walker equations:

$$\rho_1 = \phi_1 + \rho_1\phi_2 + \dots + \rho_{p-1}\phi_p$$

$$\rho_2 = \rho_{-1}\phi_1 + \phi_2 + \dots + \rho_{p-2}\phi_p$$

$$\vdots$$

$$\rho_p = \rho_{-p+1}\phi_1 + \rho_{-p+2}\phi_2 + \dots + \phi_p$$

- Note: for every **stationary AR** process values of **ACF will decay geometrically**.

Example:

$$y_t = 0.7y_{t-1} + 0.1y_{t-2} + \varepsilon_t$$

- From Yule-Walker equations:

$$\begin{cases} \rho_1 = 0.7 + \rho_1 0.1 \Rightarrow \rho_1 = 0.778 \\ \rho_2 = 0.7\rho_1 + 0.1 = 0.645 \end{cases}$$

- so

$$\rho_3 = 0.7\rho_2 + 0.1\rho_1 = 0.529$$

$$\rho_4 = 0.7\rho_3 + 0.1\rho_2 = 0.435$$

etc.

Properties of an AR(1) process

We consider an AR(1) process:

$$y_t = \phi_1 y_{t-1} + \varepsilon_t, \varepsilon_t \sim IID(0, \sigma^2)$$

- The (unconditional) mean:

$$E(y_t) = \frac{\mu}{(1 - \phi_1)}$$

- The (unconditional) variance:

$$Var(y_t) = \frac{\mu}{(1 - \phi_1^2)}$$

- ACF function:

$$\rho_k = \phi_1^k$$

Properties of MA(q)

- The expected value:

$$E(y_t) = \mu$$

- Variance:

$$Var(y_t) = \gamma_0 = (1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2)\sigma^2$$

- Covariances:

$$\gamma_k = \begin{cases} (\theta_k + \theta_{k+1}\theta_1 + \theta_{k+2}\theta_2 + \dots + \theta_q\theta_{q-k}) & \text{for } k \leq q \\ 0 & \text{for } k > q \end{cases}$$

- **A moving average process is always stationary.**
- In order to express any MA(q) process as a AR(∞) process an invertibility condition must be fulfilled:

$$\forall_i |z_i| > 1$$

where z_i are roots of a characteristic equation:

$$1 - \theta_1 z - \theta_2 z^2 - \dots - \theta_q z^q = 0$$

- Each stationary AR(p) process can be expressed as a MA(∞).

Box-Jenkins procedure

Box and Jenkins (1976) approach to estimating an ARMA model involves four steps:

- Identification
 - This involves determining the order of the model required to capture the dynamic features of the data. Graphical procedures are used (plotting the data over time and plotting the ACF and PACF) to determine the most appropriate specification.
- Estimation
 - This involves estimation of the parameters of the model specified in step 1. Depending on the model, this can be done using least squares or maximum likelihood.
- Diagnostics
 - This involves model checking - i.e. determining whether the model specified and estimated is adequate. Usually, this means examining whether the model residuals are free from autocorrelation. The ACF, PACF and Ljung-Box statistics can be used.
- Forecasting
 - This involves computing a prediction for a given period. Ex-post forecast errors can be used to assess the goodness of forecast and hence the attractiveness of a particular model.

ARIMA

AutoRegressive Integrated Moving Average model

- ARIMA modelling, as distinct from ARMA modelling, has the additional letter "I" in the acronym, standing for "integrated".
- An integrated autoregressive process is one whose characteristic equation has a root on the unit circle, i.e. is non-stationary.
- Typically researchers difference the variable as necessary and then build an ARMA model on this differenced variable.

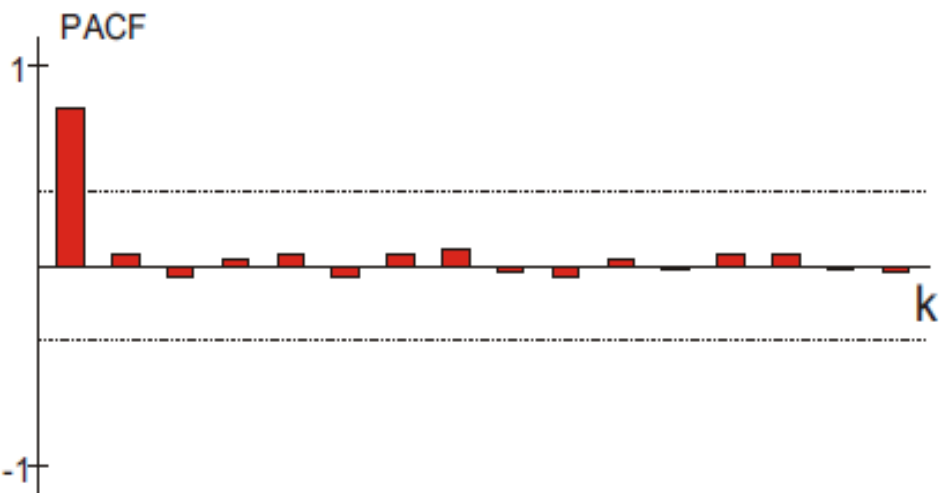
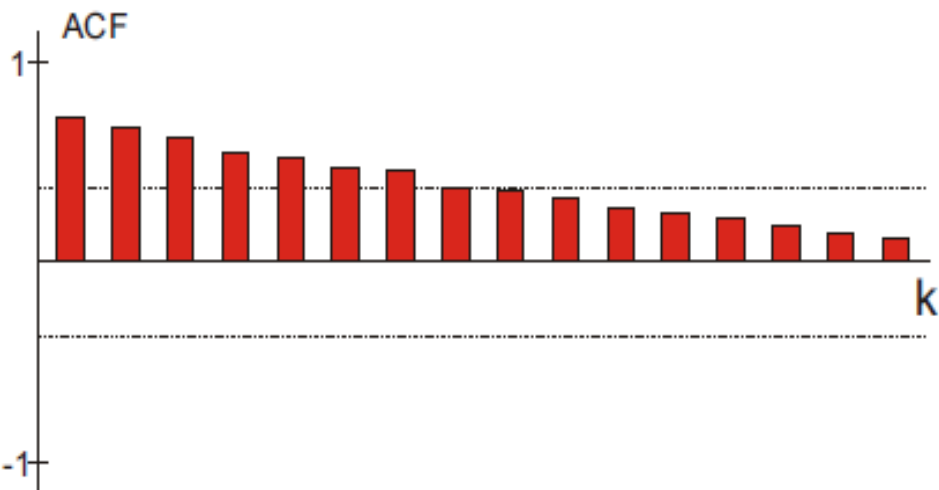
An ARMA(p,q) model on the variable differenced d times is equivalent to an ARIMA(p,d,q) model on the original data.

Identification

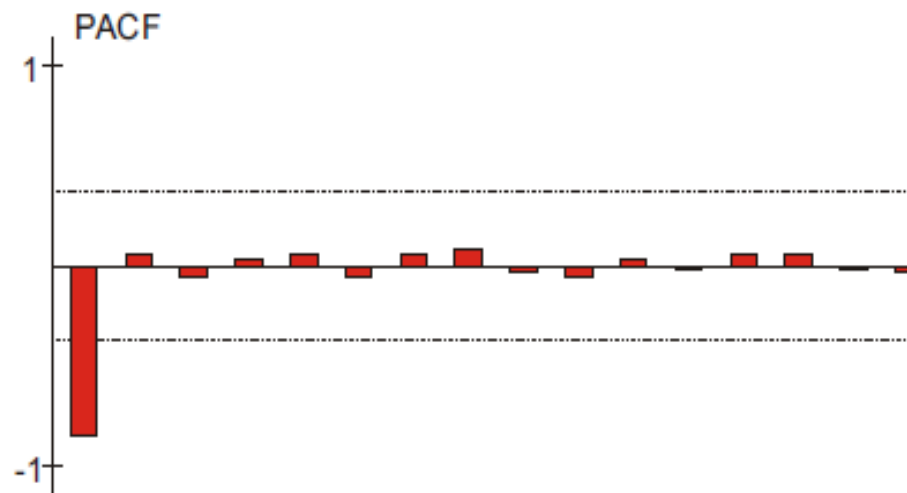
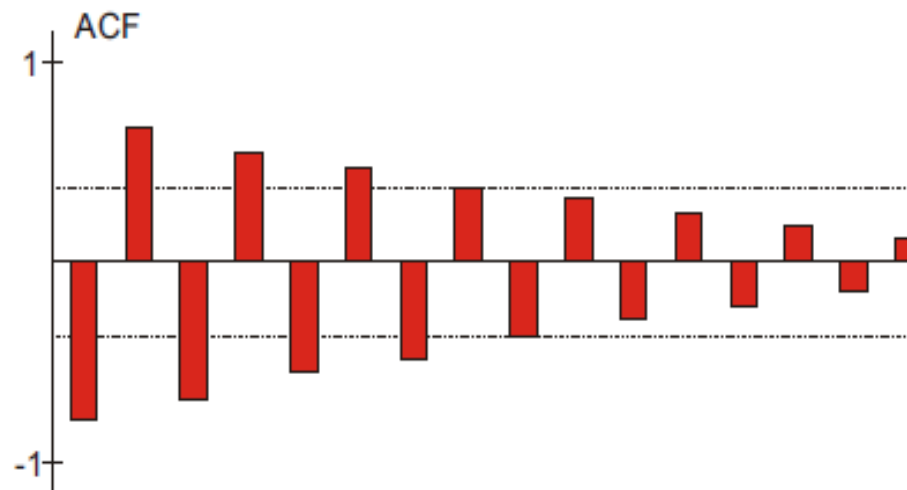
- An ARIMA(p,d,q) has three order coefficients, which should be determined before estimating the model:
 - $p \rightarrow$ order of AR process,
 - $d \rightarrow$ order of integration of an original series (Unit Root Test),
 - $q \rightarrow$ order of MA process.
- Having identified the proper order of integration d we can start identifying the orders p and q. Usually this is done upon analysing plots of ACF and PACF.

Theoretical patterns of ACF & PACF for AR(1)

$$0 < \phi_1 < 1$$



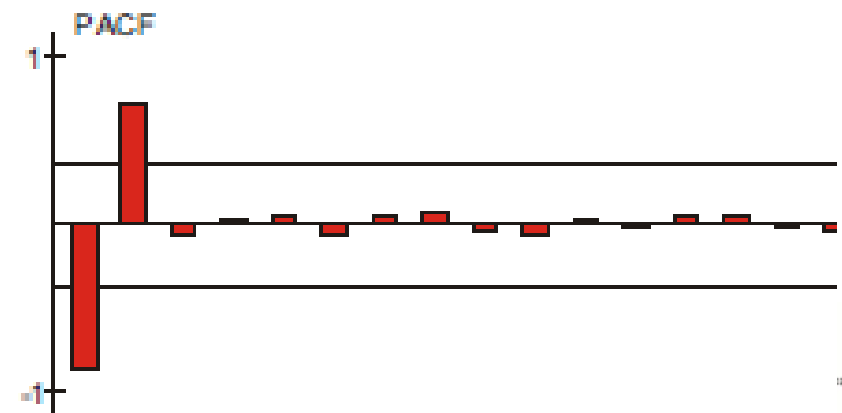
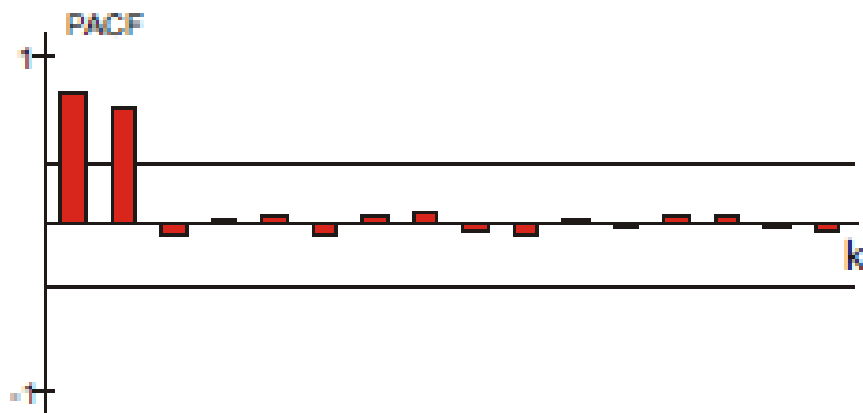
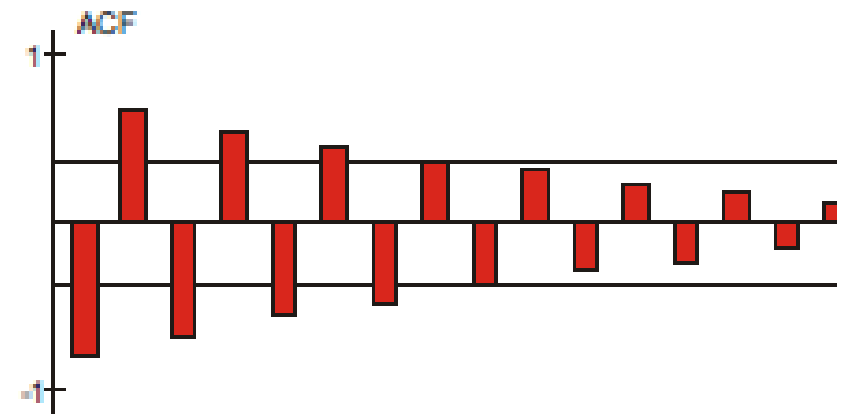
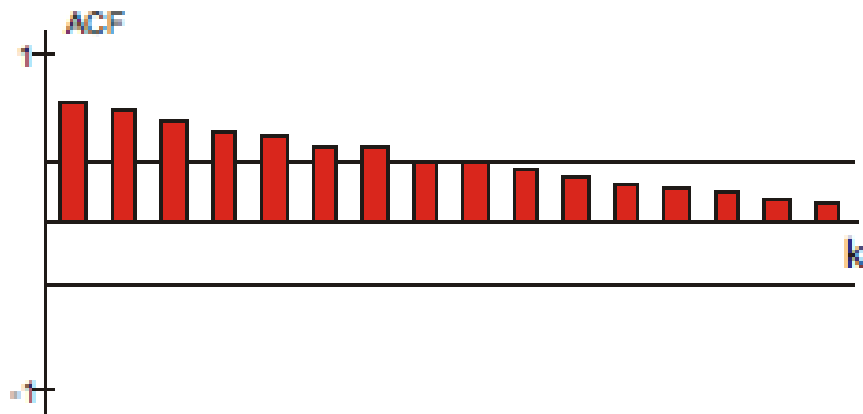
$$-1 < \phi_1 < 0$$



Theoretical patterns of ACF & PACF for AR(2) I

$$\begin{cases} 0 < \phi_1 < 1 \\ 0 < \phi_2 < 1 \\ \phi_1 + \phi_2 < 1 \end{cases}$$

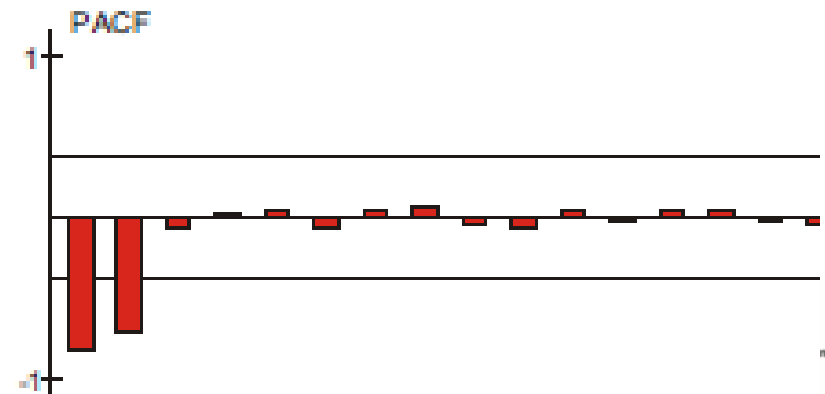
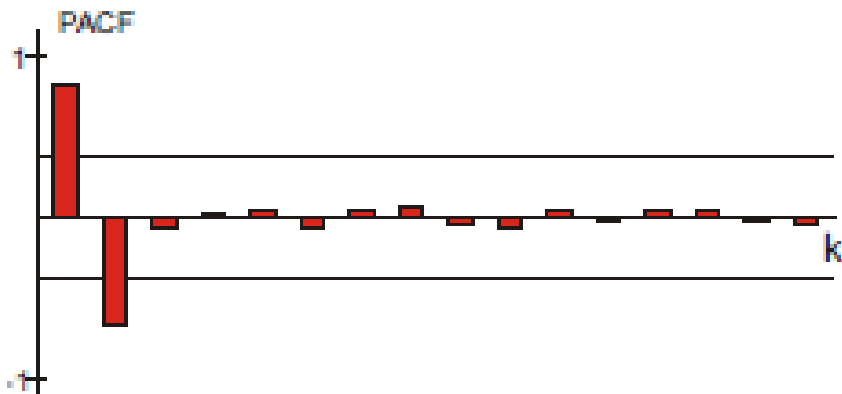
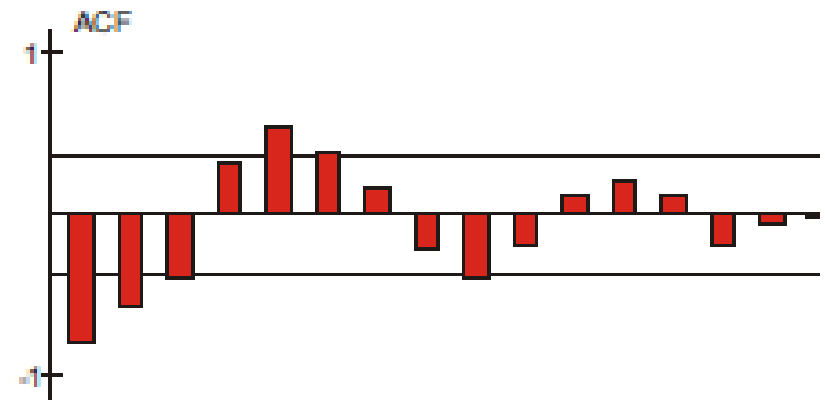
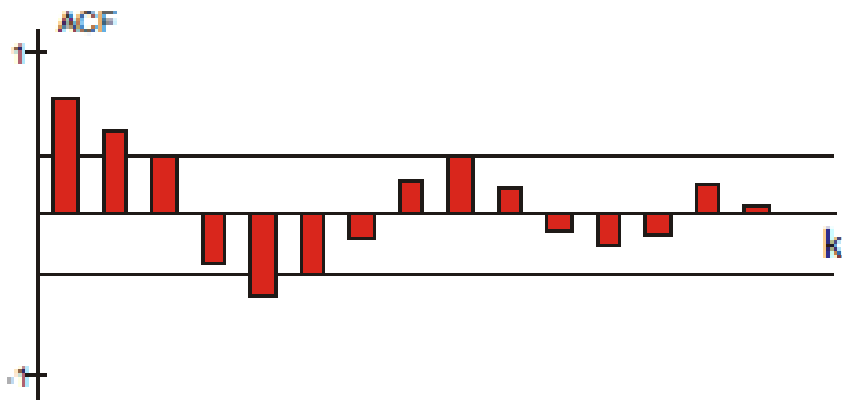
$$\begin{cases} -1 < \phi_1 < 0 \\ 0 < \phi_2 < 1 \\ \phi_1 + \phi_2 < 1 \end{cases}$$



Theoretical patterns of ACF & PACF for AR(2) II

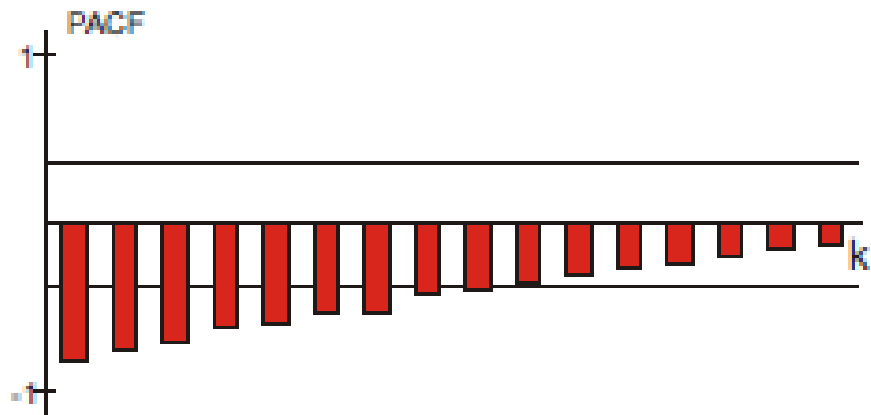
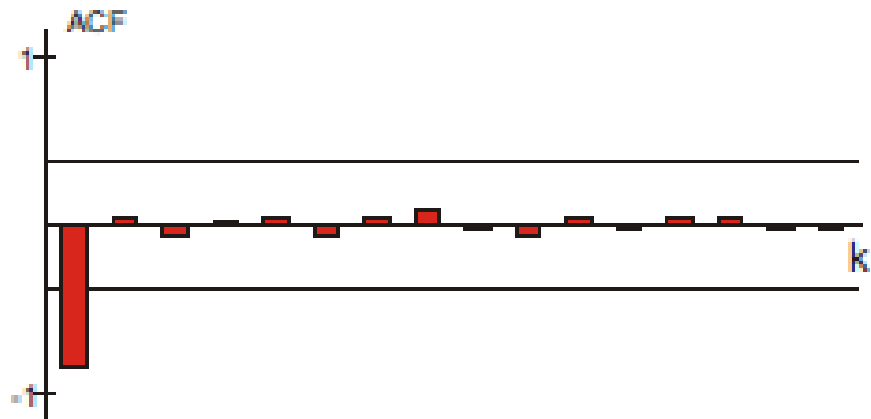
$$\begin{cases} 0 < \phi_1 < 2 \\ -1 < \phi_2 < 0 \\ \phi_1 < -\phi_1 + 1 \end{cases}$$

$$\begin{cases} -2 < \phi_1 < 0 \\ -1 < \phi_2 < 0 \\ \phi_1 < \phi_1 + 1 \end{cases}$$

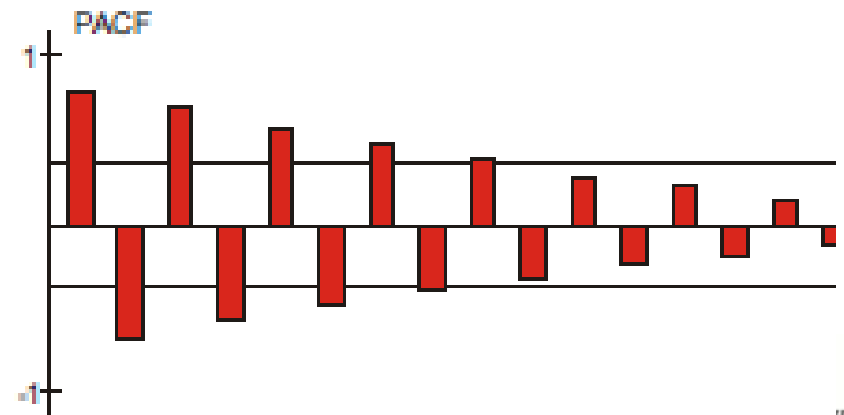
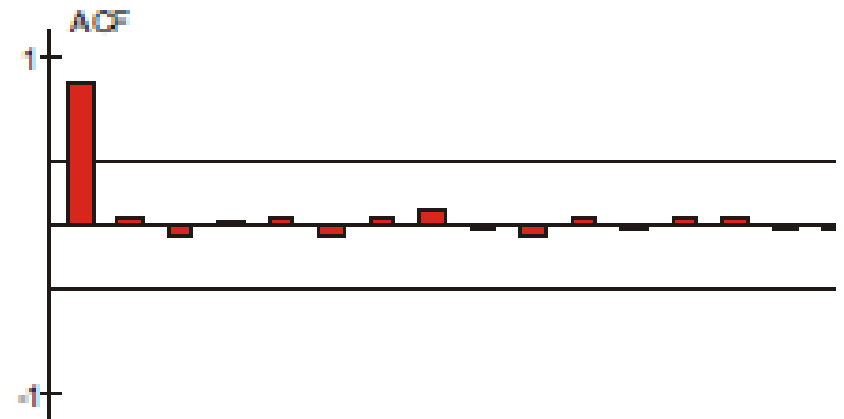


Theoretical patterns of ACF & PACF for MA(1)

$$\theta_1 < 0$$

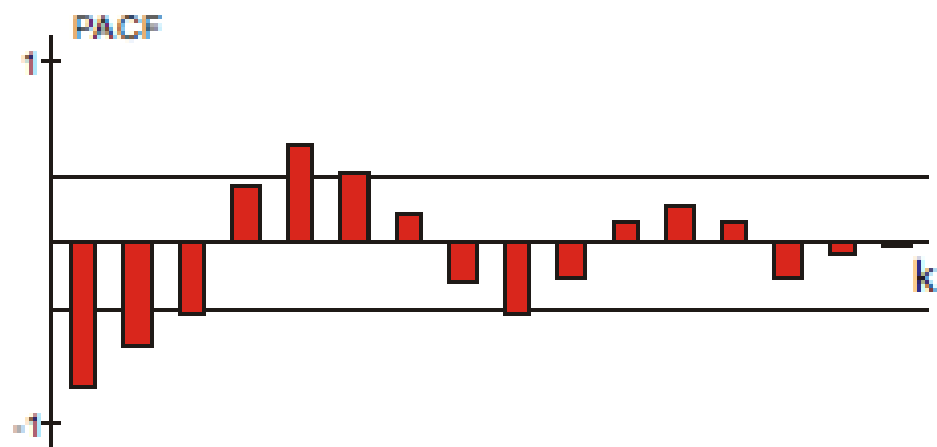
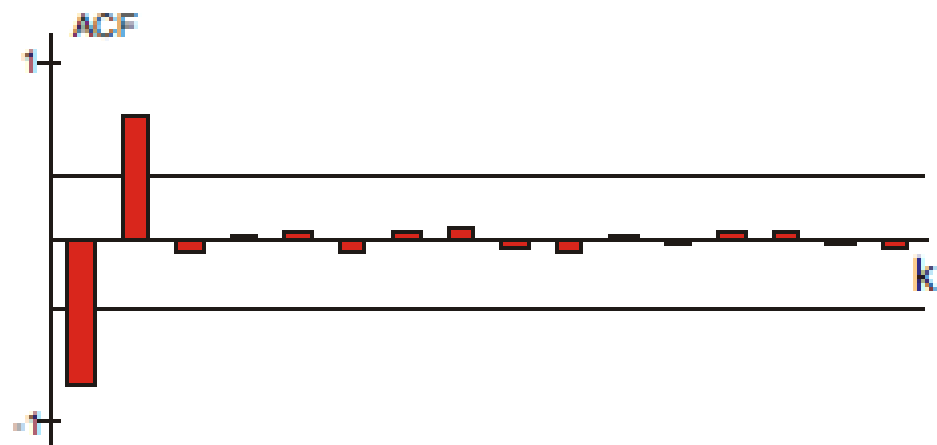


$$\theta_1 > 0$$

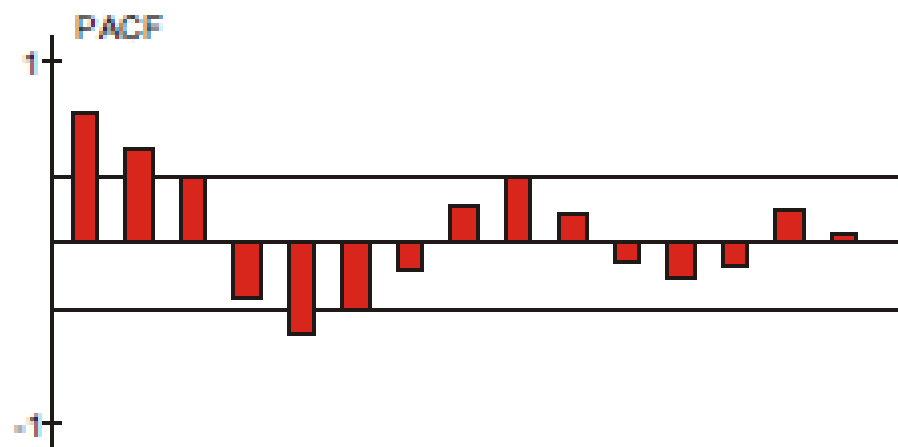
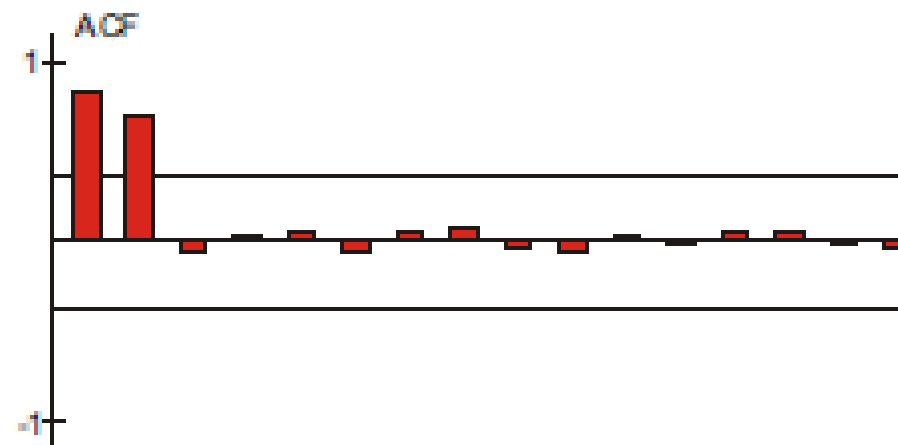


Theoretical patterns of ACF & PACF for MA(2) I

$$\begin{cases} \theta_1 < 0 \\ \theta_2 > 0 \end{cases}$$



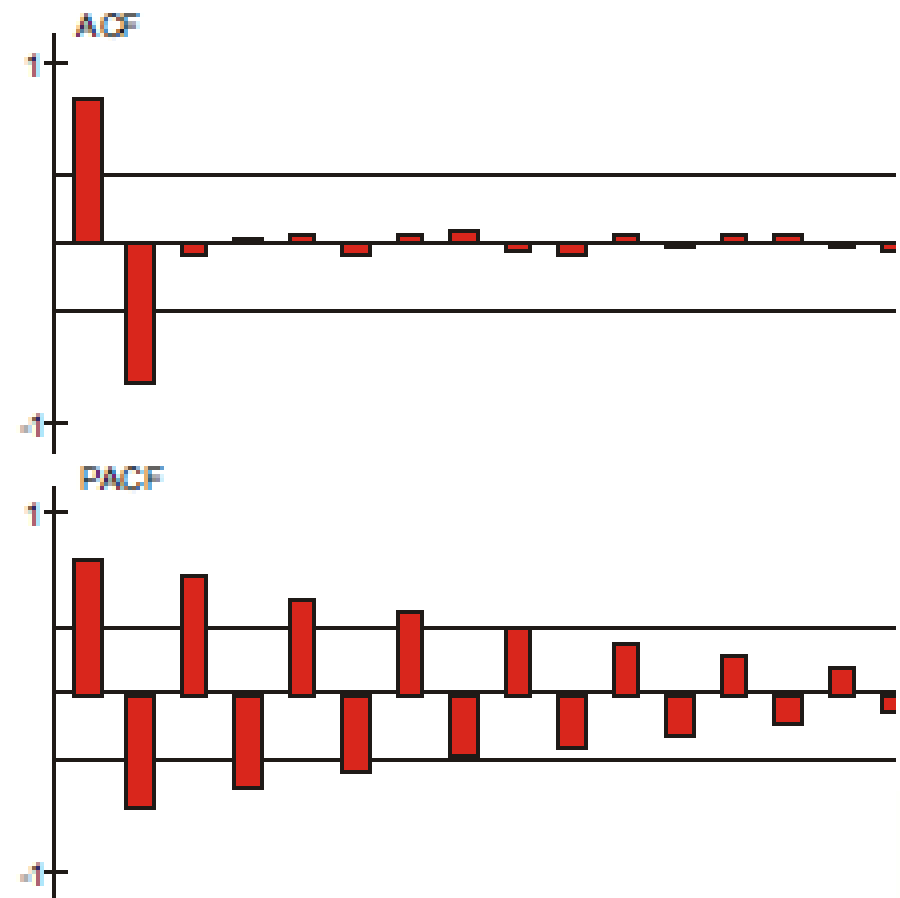
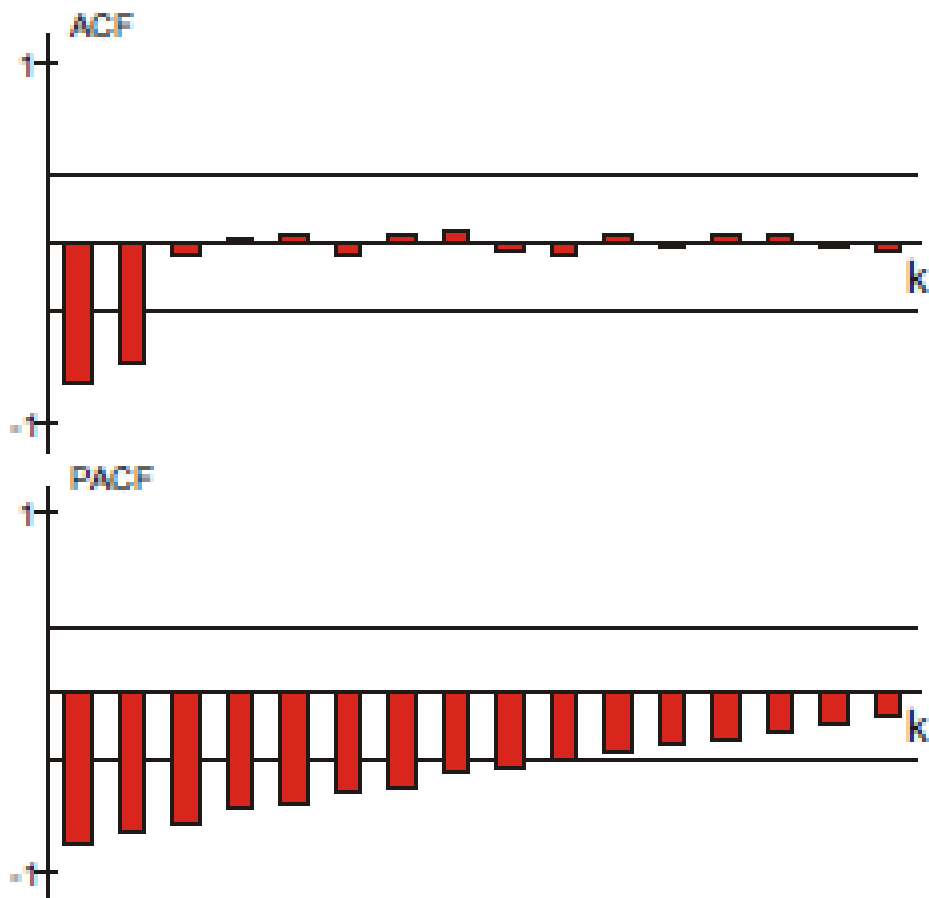
$$\begin{cases} \theta_1 > 0 \\ \theta_2 < 0 \end{cases}$$



Theoretical patterns of ACF & PACF for MA(2) II

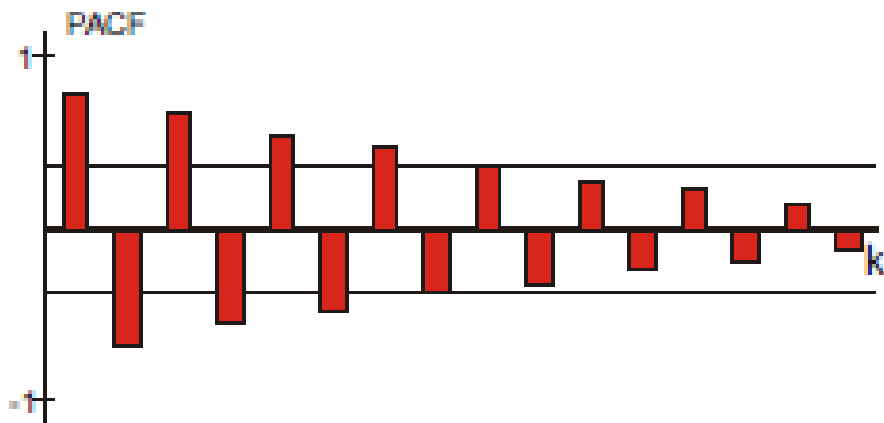
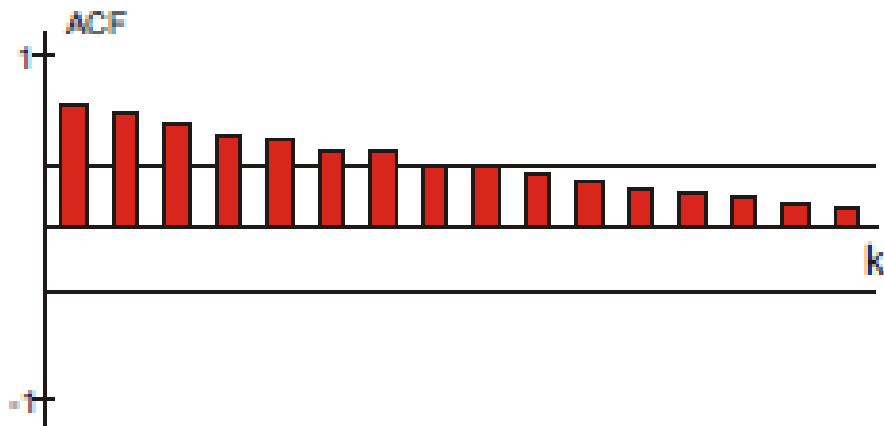
$$\begin{cases} \theta_1 < 0 \\ -1 < \theta_2 < 0 \end{cases} \quad \text{OR} \quad \begin{cases} \theta_1 > 0 \\ \theta_2 < -1 \end{cases}$$

$$\begin{cases} \theta_1 > 0 \\ -1 < \theta_2 < 0 \end{cases} \quad \text{OR} \quad \begin{cases} \theta_1 < 0 \\ \theta_2 < -1 \end{cases}$$

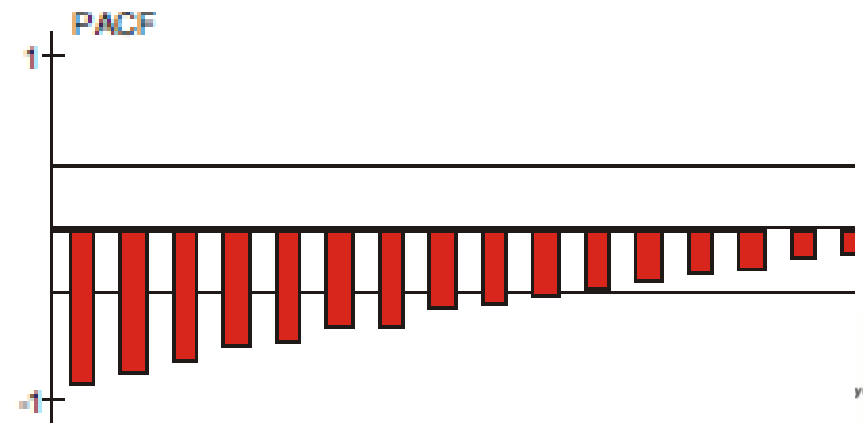
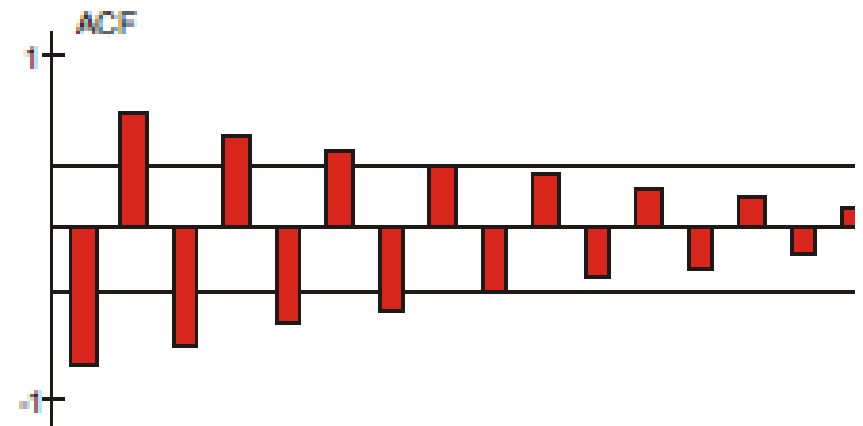


Theoretical patterns of ACF & PACF for ARMA(1,1) I

$$\begin{cases} 0 < \phi_1 < 1 \\ \theta_1 > 0 \end{cases}$$



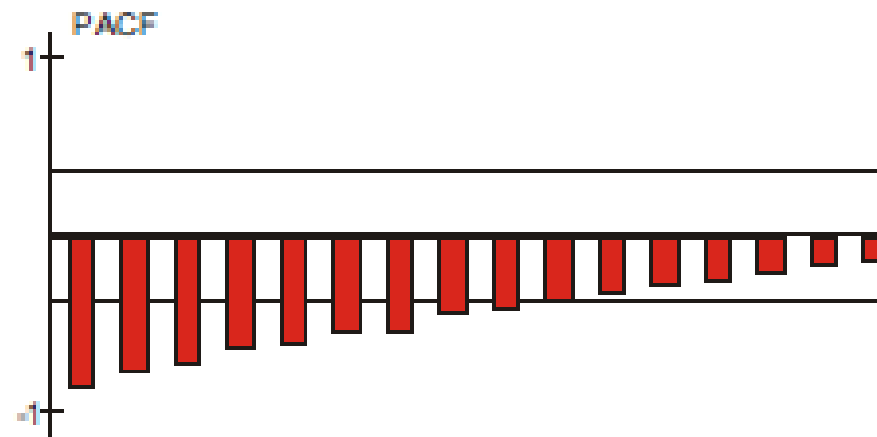
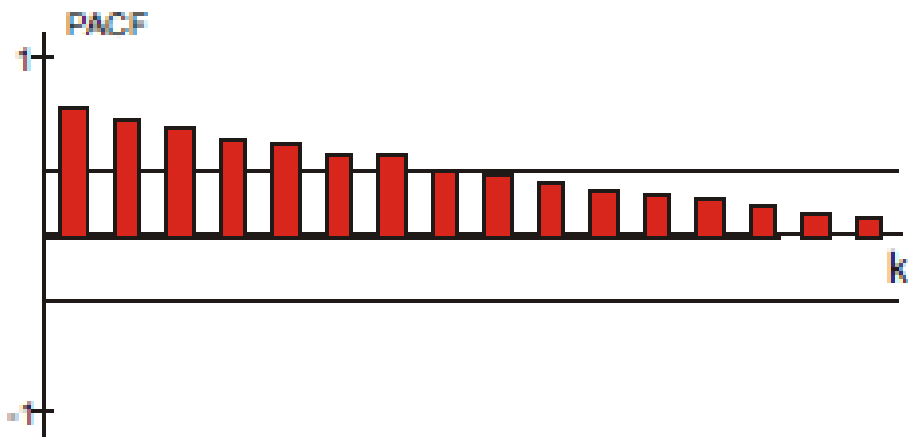
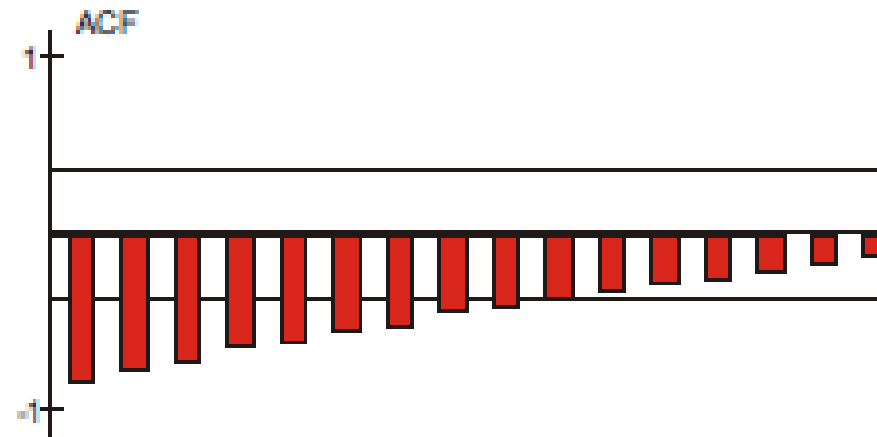
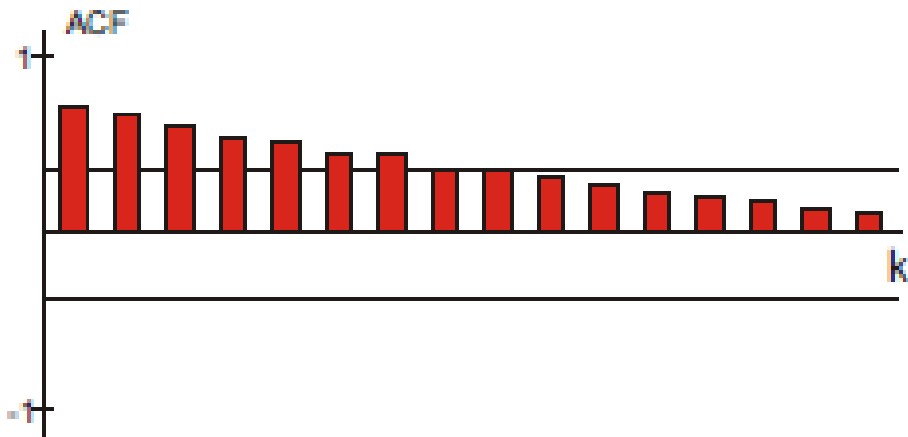
$$\begin{cases} -1 < \phi_1 < 0 \\ \theta_1 < 0 \end{cases}$$



Theoretical patterns of ACF & PACF for ARMA(1,1) II

$$\begin{cases} 0 < \phi_1 < 1 \\ \theta_1 < -1/\phi_1 \end{cases} \quad \text{OR} \quad \begin{cases} 0 < \phi_1 < 1 \\ \theta_1 < 0 \\ \theta_1 > \phi_1 \end{cases}$$

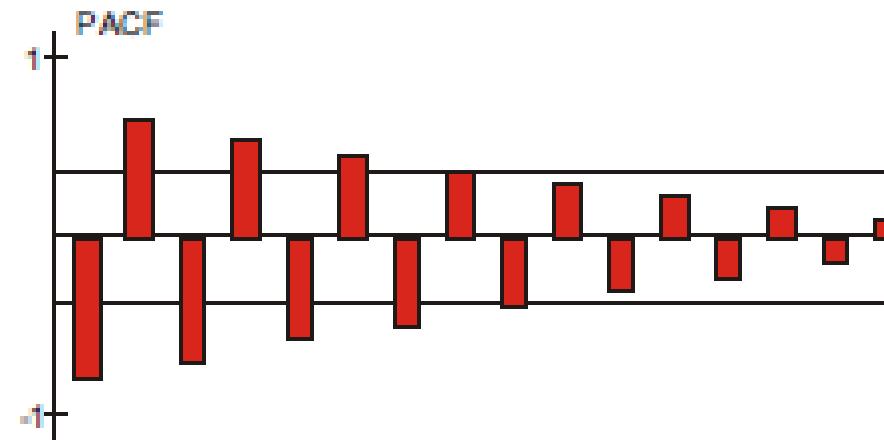
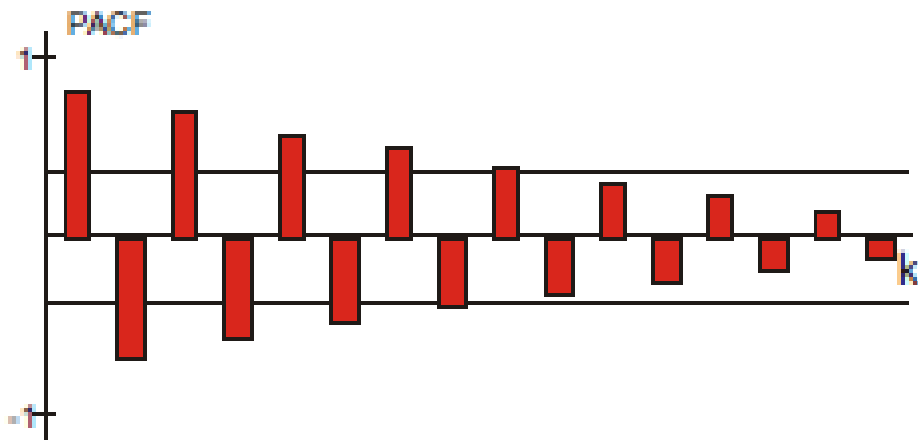
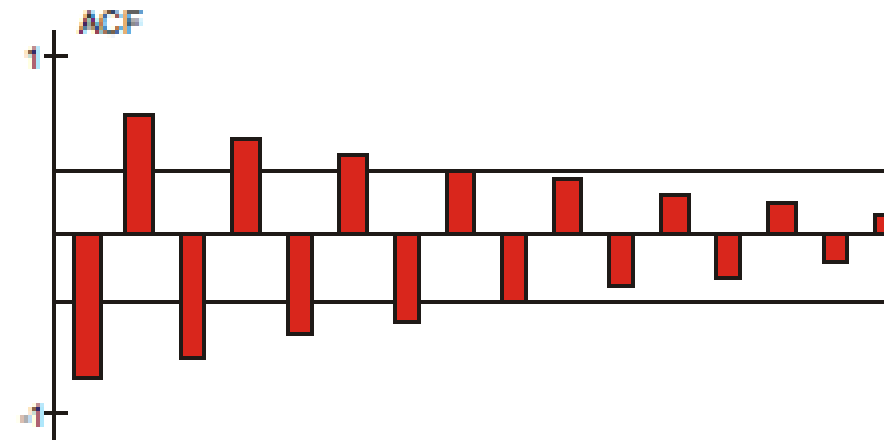
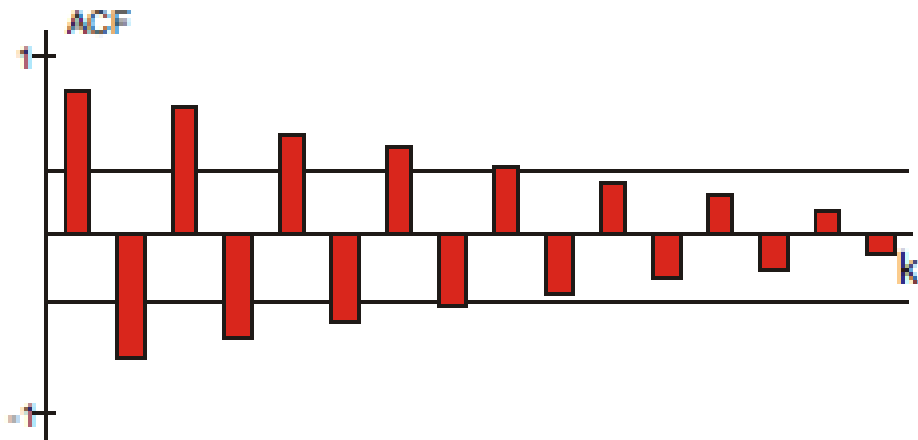
$$\begin{cases} 0 < \phi_1 < 1 \\ \theta_1 < -\phi_1 \\ \theta_1 > -1/\phi_1 \end{cases}$$



Theoretical patterns of ACF & PACF for ARMA(1,1) III

$$\begin{cases} -1 < \phi_1 < 0 \\ \theta_1 > -\phi_1 \\ \theta_1 < -1/\phi_1 \end{cases}$$

$$\begin{cases} -1 < \phi_1 < 0 \\ \theta_1 > \phi_1 \end{cases} \quad \text{OR} \quad \begin{cases} -1 < \phi_1 < 0 \\ \theta_1 > 0 \\ \theta_1 < \phi_1 \end{cases}$$



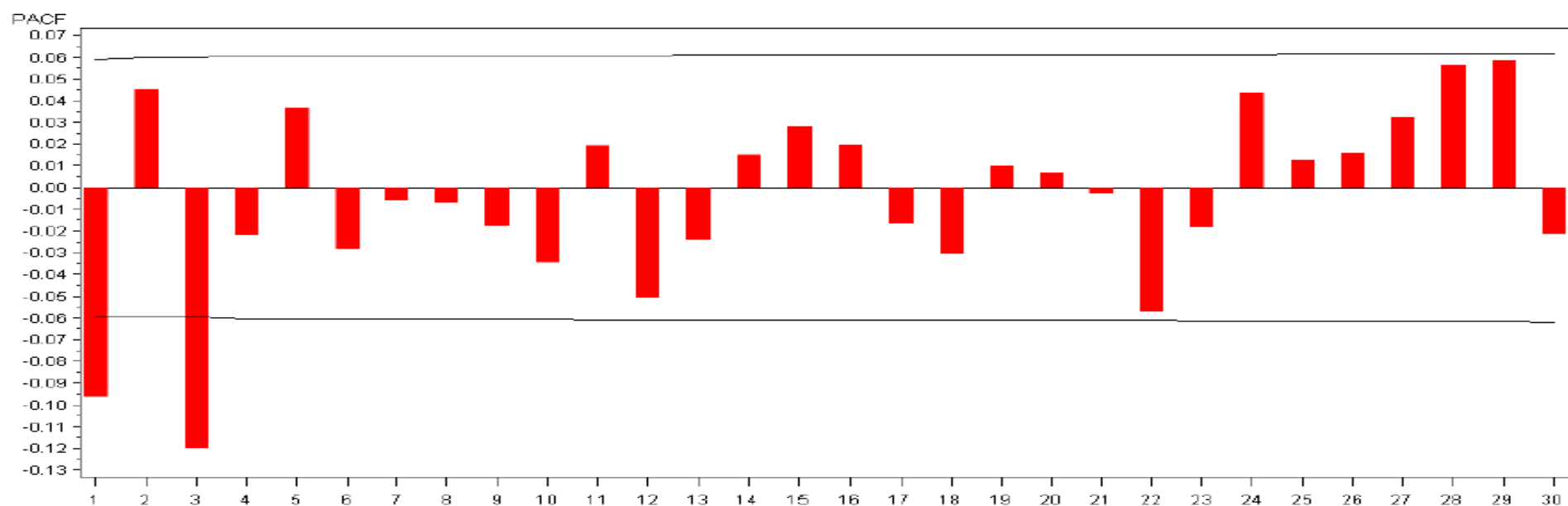
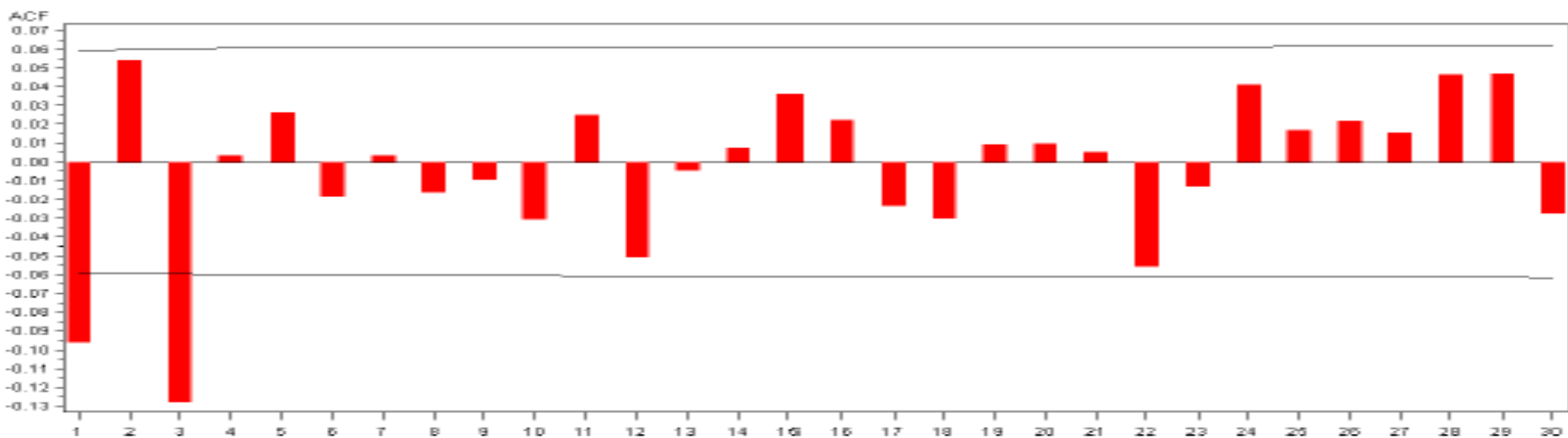
SUMMARY

- Autoregressive process - AR:
 - geometrically decaying ACF
 - last non-zero PACF $\rightarrow$ order of AR process
- Moving average process -- MA:
 - geometrically decaying PACF
 - last non-zero ACF $\rightarrow$ order of MA process
- Autoregressive moving average process - ARMA:
 - geometrically decaying ACF
 - geometrically decaying PACF

However

- When "messy" real data is used, sample ACF and PACF unfortunately rarely exhibit clear and simple theoretical patterns.
- This makes the ACF and PACF very hard to interpret, and thus it is difficult to specify a model for the data.
- Nevertheless, sample ACF and PACF plots can be used as a hint to determine what process are we actually dealing with.

Example of ACF and PACF of real data



Diagnostics

Diagnostic stage

- In a diagnostic stage we need to answer a question whether **residuals** from our model **are white noise**.
- To answer that question we can use:
 - informal way: visual inspection of ACF and PACF functions
 - formal way: Box-Pierce or/and Ljung-Box statistics testing whether k first ACF values are simultaneously equal zero.
- Distribution of ACF and PACF values

If we assume that $y_t \sim N(0,1)$ then the sample values of ACF and PACF have approximately normal distribution:

$$\hat{\rho}_k \sim N(0; \frac{1}{T})$$

Hence, the lower and upper bounds of $(1 - \alpha)$ % confidence interval for ACF and PACF values are given by:

$$\left(\Phi_{\frac{\alpha}{2}} \frac{1}{\sqrt{T}}, \Phi_{1-\frac{\alpha}{2}} \frac{1}{\sqrt{T}} \right)$$

where $\Phi(\cdot)$ denotes cumulative distribution function for standard normal distribution and T is number of observations of y_t .

Which model should we choose?

- testing whether residuals are white noise (simply analyzing ACF and PACF or using formal tests),
- the principle of parsimonious model parametrization,
- best forecast in the out-of-sample period,
- information criteria.

Information criteria

- To remove some of the subjectivity involved in interpreting the ACF and PACF, we can use **information criteria**.
- They embody two factors:
 - a term which is a function of goodness-of-fit measure,
 - and some penalty for the loss of degrees of freedom from adding extra parameters.
- So, adding a new variable or an additional lag to a model will have two competing effects on the information criteria:
 - the goodness-of-fit measure will go up, but the value of the penalty term will increase.
- The objective is to choose the number of parameters which **minimizes** the value of the information criteria.
- Three most popular information criteria are:
 - AIC - Akaike's Information Criterion:

$$AIC = \ln GOF + \frac{2K}{T}$$

- SBC - Schwarz Bayesian Criterion (also Bayesian Information Criterion, BIC)

$$BIC = \ln GOF + \frac{K}{T} \ln(T)$$

- HQIC – Hannan - Quinn criterion

$$HQIC = \ln GOF + \frac{K}{T} \ln(\ln(T))$$

where, $k=p+q+1$ is the total number of parameters estimated and T is the sample size.

Information criteria

- **Smaller criterion** value indicates **better** model fit.
- **SBC** embodies a **much stiffer penalty** term **than AIC**, while **HQIC** is somewhere **in between**.
- **SBC is strongly consistent (but inefficient)** and **AIC is not consistent, but is generally more efficient**.
- In other words, **SBC will asymptotically deliver the correct model order**, while **AIC** will on average deliver a model which is **too large**, even on a large sample of data.
- **On the other hand, the average variation in selected model orders for different samples from the same population will be greater for SBC than for AIC**.
- Overall, then, no criterion is definitely superior to others.

Forecasting with an AR model

Let us assume that we have estimated the following model:

$$y_t = \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$$

We can also write the model for future periods as:

$$y_{t+1} = \mu + \phi_1 y_t + \phi_2 y_{t-1} + \varepsilon_{t+1}$$

$$y_{t+2} = \mu + \phi_1 y_{t+1} + \phi_2 y_t + \varepsilon_{t+2}$$

$$y_{t+3} = \mu + \phi_1 y_{t+2} + \phi_2 y_{t+1} + \varepsilon_{t+3}$$

One-period ahead forecast will be given by:

$$\begin{aligned} f_{t,1} &= E(y_{t+1}|t) = E(\mu + \phi_1 y_t + \phi_2 y_{t-1} + \varepsilon_{t+1}) = \mu + \phi_1 E(y_t) + \phi_2 E(y_{t-1}) \\ &= \mu + \phi_1 y_t + \phi_2 y_{t-1} \end{aligned}$$

Two-period ahead forecast:

$$\begin{aligned} \bullet \quad f_{t,2} &= E(y_{t+2}|t) = E(\mu + \phi_1 y_{t+1} + \phi_2 y_t + \varepsilon_{t+2}) = \mu + \phi_1 E(y_{t+1}) + \phi_2 E(y_t) \\ &= \mu + \phi_1 f_{t,1} + \phi_2 y_t \end{aligned}$$

Three-period ahead forecast:

$$\begin{aligned} \bullet \quad f_{t,3} &= E(y_{t+3}|t) = E(\mu + \phi_1 y_{t+2} + \phi_2 y_{t+1} + \varepsilon_{t+3}) = \mu + \phi_1 E(y_{t+2}) + \phi_2 E(y_{t+1}) \\ &= \mu + \phi_1 f_{t,2} + \phi_2 f_{t,1} \end{aligned}$$

Forecasting with an AR model

Generally, for an AR(2) model we can write:

$$f_{t,s} = \mu + \phi_1 f_{t,s-1} + \phi_2 f_{t,s-2}$$

It can be shown that long-period forecasts will converge to the unconditional expected value of y_t .

$$f_{t,s} \xrightarrow{s \rightarrow \infty} \frac{\mu}{1 - \phi_1 - \phi_2}$$

Forecasting with an AR model

Let us assume that we have estimated the following model:

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$$y_{t+3} = \mu + \phi_1 y_{t+2} + \phi_2 y_{t+1} + \varepsilon_{t+3}$$

One-period ahead forecast will be given by:

$$\begin{aligned} f_{t,1} &= E(y_{t+1}|t) = E(\mu + \phi_1 y_t + \phi_2 y_{t-1} + \varepsilon_{t+1}) = \mu + \phi_1 E(y_t) + \phi_2 E(y_{t-1}) \\ &= \mu + \phi_1 y_t + \phi_2 y_{t-1} \end{aligned}$$

Two-period ahead forecast:

$$\begin{aligned} \bullet \quad f_{t,2} &= E(y_{t+2}|t) = E(\mu + \phi_1 y_{t+1} + \phi_2 y_t + \varepsilon_{t+2}) = \mu + \phi_1 E(y_{t+1}) + \phi_2 E(y_t) \\ &= \mu + \phi_1 f_{t,1} + \phi_2 y_t \end{aligned}$$

Three-period ahead forecast:

$$\begin{aligned} \bullet \quad f_{t,3} &= E(y_{t+3}|t) = E(\mu + \phi_1 y_{t+2} + \phi_2 y_{t+1} + \varepsilon_{t+3}) = \mu + \phi_1 E(y_{t+2}) + \phi_2 E(y_{t+1}) \\ &= \mu + \phi_1 f_{t,2} + \phi_2 f_{t,1} \end{aligned}$$

Forecasting with an MA model

Let us assume that we have estimated the following model:

$$y_t = \mu + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t$$

We can also write the model for future periods as:

$$y_{t+1} = \mu + \theta_1 \varepsilon_t + \theta_2 \varepsilon_{t-1} + \varepsilon_{t+1}$$

$$y_{t+2} = \mu + \theta_1 \varepsilon_{t+1} + \theta_2 \varepsilon_t + \varepsilon_{t+2}$$

$$y_{t+3} = \mu + \theta_1 \varepsilon_{t+2} + \theta_2 \varepsilon_{t+1} + \varepsilon_{t+3}$$

One-period ahead forecast will be given by:

$$\begin{aligned} f_{t,1} &= E(y_{t+1}|t) = E(\mu + \theta_1 \varepsilon_t + \theta_2 \varepsilon_{t-1} + \varepsilon_{t+1}) = \mu + \theta_1 E(\varepsilon_t) + \theta_2 E(\varepsilon_{t-1}) \\ &= \mu + \theta_1 \varepsilon_t + \theta_2 \varepsilon_{t-1} \end{aligned}$$

Two-period ahead forecast:

$$\begin{aligned} \bullet \quad f_{t,2} &= E(y_{t+2}|t) = E(\mu + \theta_1 \varepsilon_{t+1} + \theta_2 \varepsilon_t + \varepsilon_{t+2}) = \mu + \theta_1 E(\varepsilon_{t+1}) + \theta_2 E(\varepsilon_t) = \\ &= \mu + 0 + \theta_2 \varepsilon_t \end{aligned}$$

Three-period ahead forecast:

$$\begin{aligned} \bullet \quad f_{t,3} &= E(y_{t+3}|t) = E(\mu + \theta_1 \varepsilon_{t+2} + \theta_2 \varepsilon_{t+1} + \varepsilon_{t+3}) = \mu + \theta_1 E(\varepsilon_{t+2}) + \\ &+ \theta_2 E(\varepsilon_{t+1}) = \mu + 0 + 0 = \mu \quad \forall s \geq 3 \end{aligned}$$

Forecasting with an ARMA(p,q) model

Generally, s-period ahead forecast can be expressed as:

$$f_{t,s} = \mu + \sum_{i=1}^p \phi_i f_{t,s-i} + \sum_{j=1}^q \theta_j \varepsilon_{t+s-j}$$

where:

$$f_{t,k} = y_{t+k} \quad \text{for } k \leq 0$$
$$\varepsilon_{t+k} = \begin{cases} 0 & \text{for } k > 0 \\ \varepsilon_{t+k} & \text{for } k \leq 0 \end{cases}$$

- Hence, one can draw an important conclusion that forecasting with ARIMA(p,q) model for the number of periods **higher** than **max[p,q]** can be somewhat **questionable** since **forecasts** will **converge** to the **unconditional mean** of the dependent variable y_t .